Cyber-Physical Systems
in Mechanical Engineering TU Berlin

# PyTorch 101

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# PyTorch 101: Dataset and Forecasting Task

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**Dataset**:

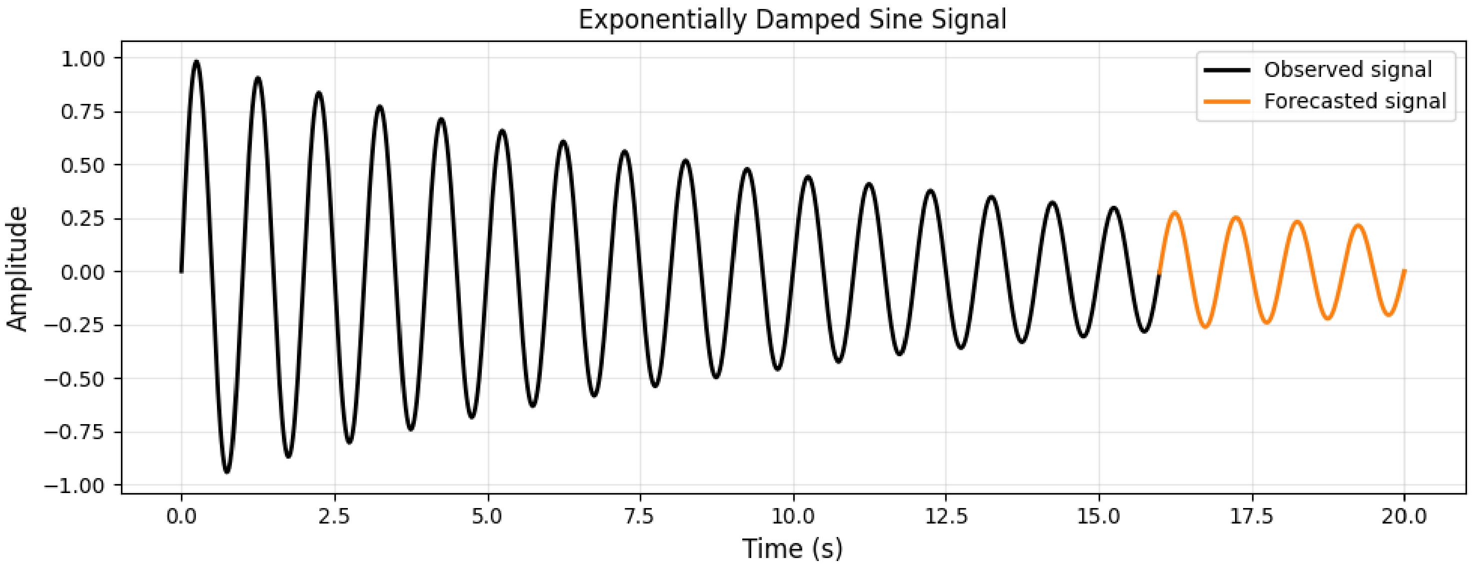
- Exponentially damped sine signal: $x(t) = A\, e^{-\lambda t} \sin(2\pi f t)$
- 20 seconds of recording

**Forecasting task:**

- Train on the first 16 seconds
- Forecast the last 4 seconds

**Preprocessing:**

- Create input-target pairs
- Normalize the dataset
- Create a custom dataset





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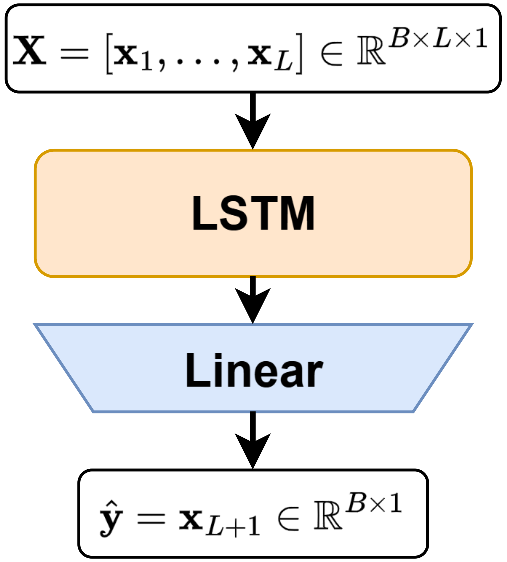
# PyTorch 101: LSTM Model Architecture

## One-Step LSTM Forecaster

- **Input:** A batch of context windows: $\boldsymbol{X} = [\boldsymbol{x}_1, \dots, \boldsymbol{x}_L] \in \mathbb{R}^{B\times L\times 1}$
- **Output:** The next value for each batch: $\hat{\boldsymbol{y}} = \boldsymbol{x}_{L+1} \in \mathbb{R}^{B\times 1\text{x}1}$





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# PyTorch 101: Training and Testing

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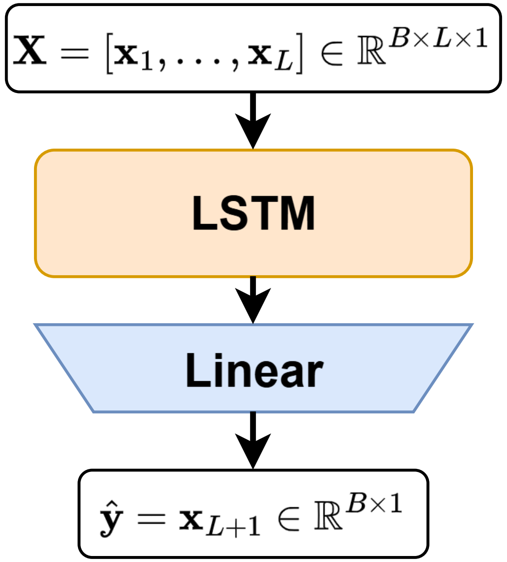
**Training:**

- Create sliding windows from the training region
- Load batches using a DataLoader
- Predict the next value
- Calculate the loss
- Update the model parameters




**Testing**

- Start with the final observed training window
- Predict the first test value
- Append the prediction to the input window
- Predict the following value
- Repeat until the complete test region is generated